Dual-Gated ALD Oxide-Channel Nonvolatile Capacitive Memory Monolithically Integrated on Foundry 40-nm CMOS for Capacitive Digital Compute-in-Memory

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***Abstract*— In this work, we experimentally demonstrate monolithic 3-D (M3D) integration of dual-gated (DG) nonvolatile capacitive memory (nvCAP) featuring an ALD W-doped In_2O_3 (IWO) channel on a foundry 40-nm CMOS chip. We introduce a novel dual-gate engineering scheme for DG nvCAP, effectively addressing the weak erase and poor retention challenges in oxide-channel ferroelectrics (FEs). The BEOL-integrated DG nvCAP on the TSMC N40 CMOS chip achieves a capacitive ON–OFF ratio (C_{ON}/C_{OFF}) of ~64.4 at V_{read} = 0 V—a record among reported BEOL-compatible nvCAPs. Furthermore, nondestructive read operation exceeding 10^9 stress cycles at $|V_{read}|$ = 1 V is demonstrated in the same device. Finally, we introduce a new "capacitive" digital compute-in-memory (Cap-DCIM) paradigm, achieving 146× higher figure-of-merit (TOPS/W × TOPS/mm^2) compared with leading analog compute-in-memories (ACIMs) and 111× lower static power versus SRAM-based DCIMs—enabled by highly scalable digital logic and BEOL integration of DG nvCAPs. We experimentally validate the operational principle of M3D Cap-DCIM by demonstrating BEOL capacitance-modulated FEOL Si transistor current amplification through monolithically integrated DG nvCAP on the TSMC N40 CMOS chip.**

***Index Terms*— CMOS + X, ferroelectric (FE), monolithic 3-D (M3D) integration, nonvolatile capacitor.**

I. Introduction

RAPIDLY increasing demand for high-performance memory and computing in artificial intelligence (AI) hardware has driven extensive research on innovative integration paradigms, such as complementary FETs (CFETs), hybrid bonding, and 2.5-D integration. Among them, CMOS + X—monolithic 3-D (M3D) BEOL integration of "X" on top of FEOL CMOS—has gained significant attention as a next-generation chip integration scheme [1]. Here, "X" refers to any memory or transistor that has BEOL process compatibility, typically defined by maintaining process temperature below <450 °C. While it was initially proposed as a theoretical concept, recent works have reported experimental demonstrations of CMOS + X with "X" such as carbon nanotube FETs (CNTFETs), resistive random access memory (RRAM), and ferroelectric random access memory (FeRAM) on FEOL CMOS in a foundry chip [1], [2], [3]. Various applications have been proposed for CMOS + X, such as gas sensor and cache with enhanced bandwidth [1], [4]. Among them, compute-in-memory (CIM) stands out as a compelling application, as its key concept—the integration of memory and compute within the same chip—naturally leverages the seamless cointegration of BEOL memory and FEOL logic in CMOS + X. Recently, nonvolatile capacitive memory (nvCAP) has been proposed as a promising memory device for charge-domain analog CIM (ACIM). ACIM based on nvCAP has been projected to deliver superior energy efficiency and compute efficiency over leading ACIM approaches based on different nonvolatile memory devices [5]. Meanwhile, among various CIM designs, digital CIM (DCIM) leveraging all-digital computation has recently emerged as a promising solution to address the fundamental challenges with ACIM, suffering from process/voltage/temperature (PVT) variations and scalability challenges in advanced nodes [Fig. 1(b)] [2], [3]. Despite its promising advantages, DCIM is also known to have several drawbacks, such as large static power from the SRAM memory and bulky cell area [6].

In this work, we experimentally demonstrate a dual-gated (DG) nvCAP with an ALD oxide channel (DG nvCAP)—monolithically integrated on a TSMC N40 CMOS chip—which enables a novel DCIM approach addressing the limitations of both the conventional SRAM-based 10T DCIM [7] and ACIM. First, we develop a novel dual-gate engineering scheme for DG nvCAP, mitigating the inherent weak erase and poor retention issues in oxide-channel ferroelectrics (FEs).

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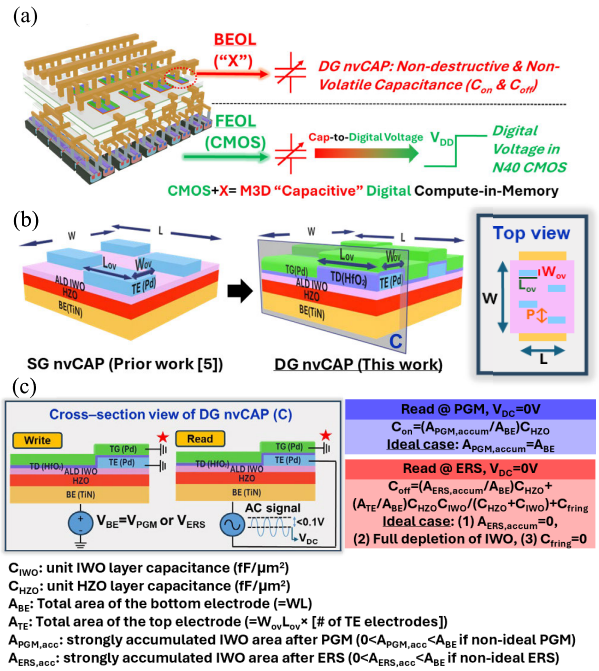


Fig. 1. (a) Schematic overview of the CMOS + X and its application proposed in this work. (b) Illustration of the DG nvCAP proposed in this work. (c) Cross section of the DG nvCAP, and the description of its operational principle.

Next, we experimentally demonstrate the CMOS + X flow for monolithically integrating DG nvCAP on the BEOL of TSMC N40 CMOS chip. The optimized BEOL-integrated DG nvCAP on TSMC N40 CMOS achieves a capacitive ON–OFF ratio (C_{on}/C_{off}) of 64.4 with nondestructive read operation (NDRO) of $>10^9$ stress cycles at accelerated read condition $|V_{read}| = 1$ V. Finally, we propose a new M3D nondestructive "capacitive" DCIM (Cap-DCIM), which combines the scalability of DCIM and nondestructive and nonvolatile nature of BEOL-integrated nvCAP [Fig. 1(a)]. We experimentally demonstrate its key operational principle, capacitance-modulated FEOL Si transistor current amplification by monolithically integrating a 2T-1C test structure consisting of the BEOL-integrated DG nvCAP and FEOL TSMC N40 core/IO transistors. Based on the measured data, we develop a cell-level circuit model for Cap-DCIM, showing close agreement with the experimental results. The cell-level and macrolevel PPA analyses conducted with this validated circuit model suggest the strong promise of Cap-DCIM with technology scaling, particularly for read-intensive applications. This article is an extension of the conference paper [8] with detailed discussions on the DG nvCAP operation mechanism and array-level operation of Cap-DCIM.

II. Operational Principle of ALD Oxide-Channel nvCAP on Si Wafer

We outline the general operational principle of an ALD oxide-channel FE nvCAP applicable to all the designs presented in this work [Fig. 1(b) and (c)]. Oxide-channel FE nvCAP replaces the Si channel of the conventional FE nvCAPs with an amorphous oxide semiconductor channel [5]. It should be noted that all the capacitance expressions (C_{on}, C_{off}) used throughout this article are in normalized units (fF/μm^2) unless otherwise explicitly specified in absolute units (fF). During write, the FE dipoles are aligned to up (down) direction through a PGM (ERS) operation by applying $V_{PGM} > 0$ ($V_{ERS} < 0$) across bottom electrode (BE) and top electrode (TE). At the ON-state ($V_{DC} = 0$ V after PGM), the upward FE dipoles lead to channel accumulation. Thus, the measured ON-state capacitance during read, C_{on}, can be expressed as ($A_{PGM,acc}/A_{BE}$)C_{HZO}, where $A_{PGM,acc}$ is the accumulated channel area. $A_{PGM,acc} < A_{BE}$ can be caused by incomplete PGM due to the channel *RC* delay, resulting in weak accumulation in the nonoverlap region [5], [9]. At the OFF-state ($V_{DC} = 0$ V after ERS), the downward FE dipoles lead to channel depletion. Thus, the measured OFF-state capacitance during read, C_{off}, is expressed as the parallel connection of three components: 1) ($A_{ERS,acc}/A_{BE}$)C_{HZO} (representing the nonideal ERS); 2) TE–BE overlap capacitance; and 3) TE–BE fringing capacitance (C_{fring}). It should be noted that $A_{ERS,acc} > 0$ can be caused due to the incomplete ERS (leading to partially accumulated channel) [9]. TE–BE overlap capacitance is the serial connection of (A_{TE}/A_{BE})C_{IWO} and (A_{TE}/A_{BE})C_{HZO}. C_{fring} originates from the fringing field between TE and BE.

III. DG ALD Oxide-Channel nvCAP

In this work, we propose DG nvCAP with an ALD oxide-channel as the "X" to be integrated on FEOL CMOS [Fig. 1(b)]. Prior work introducing an oxide-channel FE nvCAP has adopted a single-gated nvCAP (SG nvCAP) design [5]. In [5], C_{on}/C_{off} comparable to that of the Si baseline has been achieved by means of W-doping optimization (optimized at 4 at.%) and overlap area engineering. However, oxide-channel FEs including SG nvCAP suffer from limited retention and weak erase issue due to the inherent positive charge deficiency of oxide materials [9], [10], [11]. This issue is mitigated in the DG nvCAP by introducing the top gate (TG), separated from TE by a thin ~4.5-nm top dielectric (TD) layer on top of the reported SG nvCAP [9] (the detailed process flow for SG/DG nvCAP studied in this section will be described in Section IV with the CMOS + X flow). The major change in the operation compared with the SG nvCAP is that TG is grounded during the read and write operations, while C_{on} and C_{off} of the DG nvCAP are taken between TE and BE [Fig. 1(c)].

There are three key device-level technical advancements achieved in DG nvCAP. We demonstrate the advancements through the comparison between SG [9] and DG nvCAP characteristics on Si wafers. First, the retention of C_{on} can be improved. Prior work reported unstable retention of the low V_{th} state of the W-doped In2O3 (IWO) channel FeFETs with the same W concentration (4 at.%) [10]. Similarly, SG nvCAP suffers from poor C_{on} retention (~10^3 s at 25 °C) [5], [9]. The main problem of SG nvCAPs is that the nonoverlap region suffers from weak channel accumulation due to the *RC* delay along the nonoverlap channel region during PGM, resulting in ineffective screening of the polarization bound charges (Fig. 2). Consequently, high depolarization of the FE layer in the nonoverlap leads to poor C_{on} retention. In DG nvCAPs, TG provides additional screening charges for the nonoverlap region with weak channel accumulation, reducing the depolarization field [Figs. 2 and 3(a)]. Such effects are experimentally demonstrated in Fig. 3(b), showing substantial

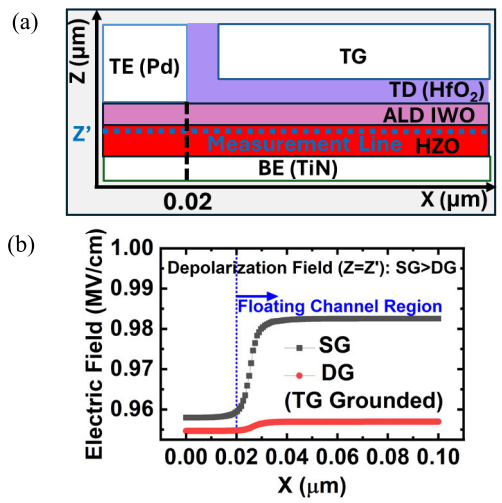


Fig. 2. (a) Simulated DG nvCAP structure in Sentaurus TCAD. (b) Simulated electric field (depolarization field) at the ON-state as a function of lateral distance from the overlap region.

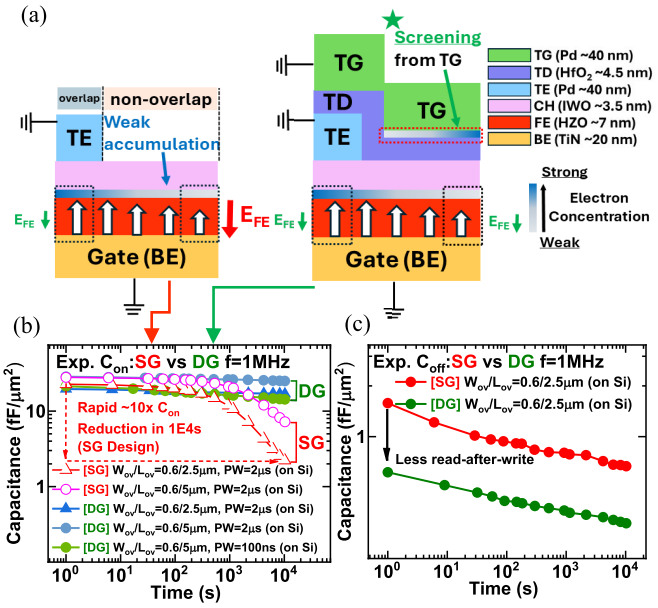


Fig. 3. (a) Illustration of the retention enhancement mechanism in DG nvCAP. (b) C_{ON} and (c) C_{OFF} retention results of DG and SG designs measured on Si wafer under varying PGM PWs highlighting the significant retention improvement using DG design. 20 wakeup cycles of (2.5 V/1 s, −4.5 V/1 s) followed by a reset ERS: −4.5 V/400 μs (PGM: 2.5 V/PW μs) were applied prior to applying PGM (ERS) pulses for C_{ON} (C_{OFF}) retention test. W/L/P = 20/10/5 μm.

C_{on} retention enhancement in DG nvCAPs (>10^4 s at 25 °C) over SG nvCAPs (~10^3 s at 25 °C).

Second, the weak erase issue [9], [11], the fundamental challenge in oxide-channel FEs can be mitigated. SG nvCAPs suffer from weak erase, mainly due to the lack of positive charges in the nonoverlap region of the channel that supports the negative voltage drop across HZO. Grounded TG provides positive charges (holes) for the nonoverlap region, leading to a larger negative voltage drop across HZO during ERS. This is evidenced by the improved read-after-write-delay of C_{off} in DG nvCAPs over SG nvCAPs in Fig. 3(c) [9]. Furthermore,

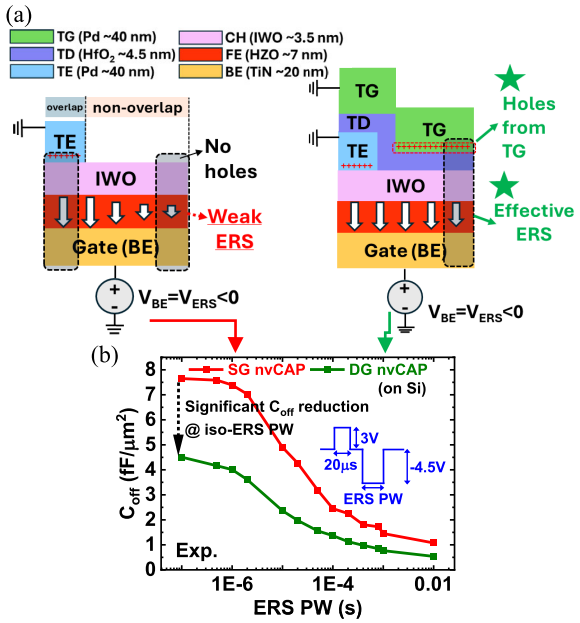


Fig. 4. (a) Illustration of the physical mechanism of the improved ERS efficiency in DG nvCAP. (b) Comparison of pulse-mode ERS between DG nvCAP and SG nvCAP on Si wafer. The geometry condition for nvCAPs is W/L/W_{ov}/L_{ov}/P = 20/10/0.6/2.5/5 μm.

the improved pulse-mode ERS efficiency in [Fig. 4(a) and (b)] suggests the effect of grounded TG.

Finally, C_{off} can be suppressed by the reduction of the fringing field between TE and BE. Fig. 5(a) and (b) shows the fringing capacitance component (C_{fring}) originates from the fringing field between TE and BE. It is expected that C_{fring} becomes an increasingly dominant component of C_{off} with nvCAP area scaling, since C_{fring} does not scale with the TE area unlike TE–BE overlap capacitance. Furthermore, in practical operation, $A_{ERS,acc}$ [Fig. 2(c)] and C_{fring} could gradually increase with repeated bipolar write cycling due to up-polarized FE domain pinning and incomplete ERS operation resulting in a more conductive channel at ERS. By introducing the grounded TG in the DG design, the TG intercepts the fringing field from the BE that would otherwise be shielded by the charges from TE. This leads to substantial C_{off} reduction, as the capacitance in DG nvCAP is defined between TE and BE. The experimental plots in Fig. 5(c)[1] indicate the effect of grounded TG in reducing C_{off} and enhancing C_{on}/C_{off}, especially after repeated bipolar write cycling through dual DC C–V sweeps.

IV. CMOS + X Flow: M3D Integration of DG nvCAP on TSMC N40 CMOS Chip

The CMOS + X flow for the M3D integration of DG nvCAP on the TSMC N40 CMOS chip and the process flow for the baseline SG/DG nvCAPs on Si wafers are comprehensively shown in Fig. 6(a)–(c). In the CMOS + X

[1]We would like to point out a typographical issue in Fig. 5(a) of our conference version that was corrected in Fig. 5(c). One cycle defined in Fig. 5(a) of the conference paper consisted of five subcycles, so the number of bipolar write cycling performed before C–V measurement in Fig. 5(c) is 95.

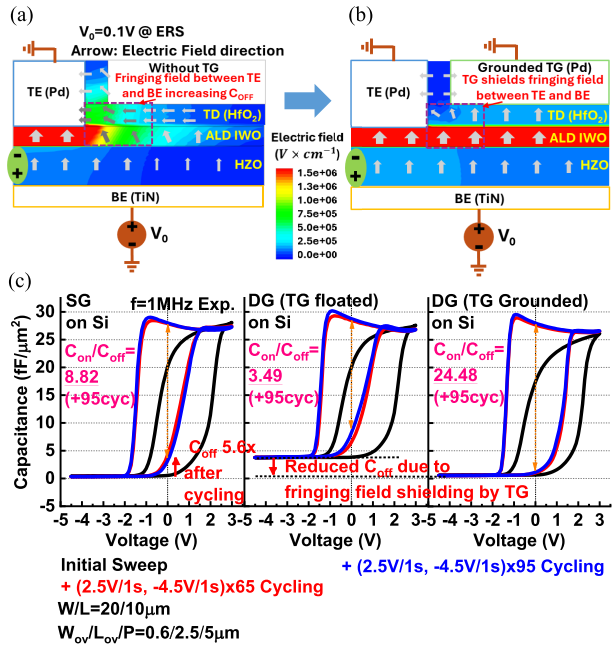


Fig. 5. Simulated electric field of (a) SG nvCAP and (b) DG nvCAP in Sentaurus TCAD. (c) Comparison of the measured *C*–*V* curves of SG nvCAP, DG nvCAP (TG floating), and DG nvCAP (TG grounded). The thickness of each material layer in the SG nvCAP and DG nvCAP is the same as that in Figs. 3 and 4.

flow, the foundry process from TSMC is stopped midway through BEOL metallization, followed by top passivation layer deposition and planarization. In our university cleanroom, the 12-in wafer from TSMC is diced into ~1 × 1 cm sized chips [Fig. 6(b)]. After that, we carry out BEOL integration of the DG nvCAPs on the top metal layer of the diced chips.

BEOL integration of DG nvCAP begins by removing the passivation layer deposited on top metal layer dielectric. The BE and interconnects between FEOL transistor pads and nvCAP BE pads are then deposited and patterned. Subsequently, a 7-nm HZO layer followed by a ~130-nm W sacrificial layer is deposited. Crystallization annealing is then performed at 450 °C for 15 min. After annealing, ~130-nm W is removed using diluted KOH solution. Next, ~3.5-nm-thick IWO with 4% W concentration is deposited with ALD supercycles and patterned. After depositing and patterning ~40-nm Pd serving as the TE and BE pads, a ~4.5-nm HfO_2 TD is deposited. Finally, ~40-nm Pd is deposited and patterned for TG. Fig. 6(c) shows the SEM top-view and TEM cross section of the BEOL-integrated DG nvCAP and the 2T-1C test structure for Cap-DCIM on TSMC N40 CMOS chip. It should be noted that the I_d–V_{gs} characteristics of the FEOL transistors after the optimized CMOS + X process (with max process temperature 450 °C) show minimal shift from the pristine transistors and exhibit good agreement with the TSMC provided device model [8]. Furthermore, the achieved $2P_r$ (~ 34.6 μC/cm^2) of the FE layer (7-nm HZO) on the TSMC N40 CMOS chip is comparable to that on the baseline Si wafers [8].

V. Device Characteristics of the Optimized DG nvCAP After M3D Integration

We optimize the C_{on}/C_{off} ratio of the BEOL-integrated DG nvCAP. Prior work suggests scaling of the width of the TE (W_{ov}) is an effective knob for C_{on}/C_{off} enhancement [5]. We scale W_{ov} down to 330 nm, achieving C_{on}/C_{off} of ~64.4 on TSMC N40 CMOS chip (Fig. 7(a)[2]). The rest of the geometry parameters are L_{ov} = 10 μm, and W/L/P = 20/10/5 μm.

Furthermore, the reliability characteristics (retention, read endurance, write endurance) of the optimized DG nvCAP on TSMC N40 CMOS chips are examined. Before each reliability test, 40 wakeup cycles of (3 V/1 s, −4.5 V/1 s) were applied to the BE of DG nvCAP while TG and TE were grounded. Fig. 7(b) shows that the retention of C_{on} and C_{off} at 25 °C is >10^4 s. However, the retention of C_{on} at 85 °C is limited to ~10^3 s. C_{on} retention loss at elevated temperature can be attributed to pronounced imprint, degraded P_r with increasing temperature [12], [13], [14]. Moreover, the large device area (200 μm^2) is another critical factor that leads to poorer retention at elevated temperatures. It is expected that the scaling of the device area will naturally lead to a lower depolarization across nonoverlap regions, which is supported by Fig. 2 and [9]. In this sense, it will be of interest in future work to develop on-chip measurement schemes to test the retention of scaled devices (below μm-scale) with aF-level capacitance at elevated temperatures (>85 °C).

The read-endurance characteristics of the optimized DG nvCAP are shown in Fig. 7(c). Read-stress cycles with accelerated condition $|V_{read}|$ = 1 V are applied to the device, and the AC small-signal capacitance is measured at designated cycle intervals with a 1-MHz, 0.1-V amplitude signal. The result shows minimal degradation of C_{on} and C_{off} states over 10^9 read-stress cycles. Moreover, the device maintains a measurable capacitance memory window up to approximately 2.4×10^5 write-endurance cycles, after which dielectric breakdown occurs [Fig. 7(d)]. Prior work [15] has identified the oxygen vacancy generation at the channel/FE layer interface and FE bulk as the primary physical origin of endurance failure in oxide-channel FeFETs. Thus, channel/FE interface and FE bulk engineering can be performed to further enhance the write endurance performance of DG nvCAPs in future work. The benchmark of BEOL-compatible nvCAPs is shown in Table I, highlighting the record C_{on}/C_{off} ratio among the reported BEOL-compatible nvCAPs and the first BEOL-"integrated" nvCAP achieved in this work [16], [17].

VI. Proposal, Experimental Validation, and Modeling of Cap-DCIM

A. Proposal of Cap-DCIM

We propose that a BEOL-integrated DG nvCAP can be connected with FEOL transistors to form a 3T-1C unit cell for Cap-DCIM, a new computing paradigm resolving the critical drawbacks of the conventional SRAM-based 10T DCIM [7]

[2]We would like to point out a typographical issue in Fig. 5(d) of our conference paper that was corrected in Fig. 7(a). The number of bipolar write cycling performed before *C*–*V* measurement is 95, as the one cycle defined in Fig. 5(d) of the conference paper consisted of five subcycles.

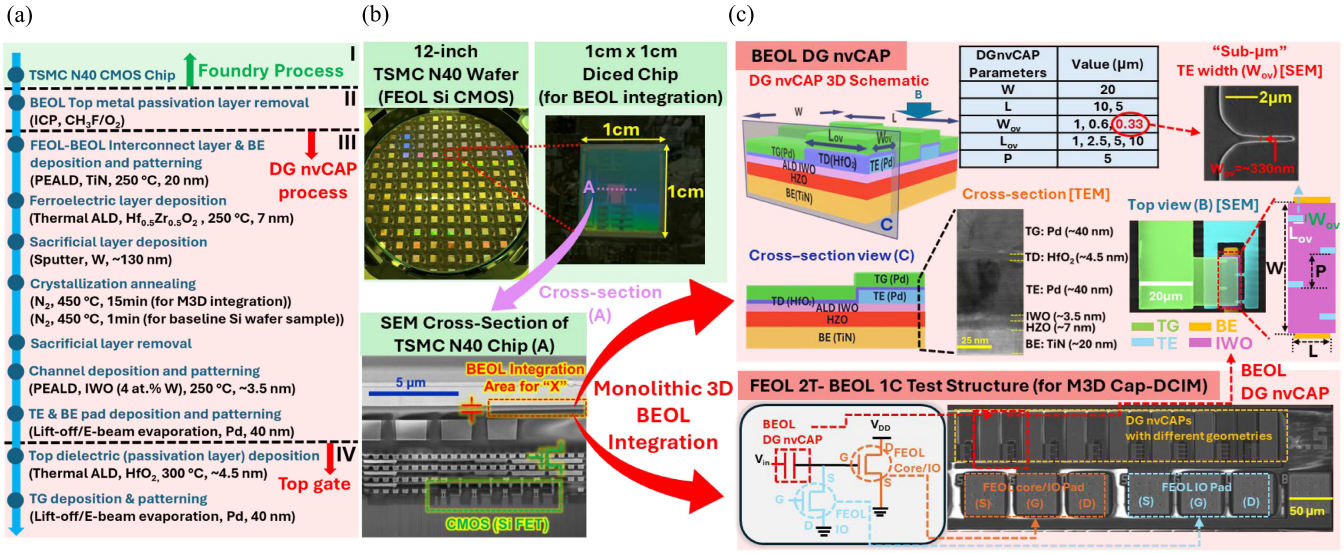


Fig. 6. (a) Outline of the M3D integration (CMOS + X) flow (I–IV) for DG nvCAP on TSMC N40 CMOS chips. Baseline SG nvCAP and DG nvCAP on Si wafers are fabricated using the process flow III, and III and IV, respectively. (b) Overview of the CMOS + X integration, including the 12-in wafer photograph, diced chip photograph, and SEM cross-sectional image of the TSMC N40 CMOS chip. (c) Description of the BEOL-integrated DG nvCAP and the 2T-1C test structures for Cap-DCIM on TSMC N40 CMOS, including the SEM top-view and TEM cross section images.

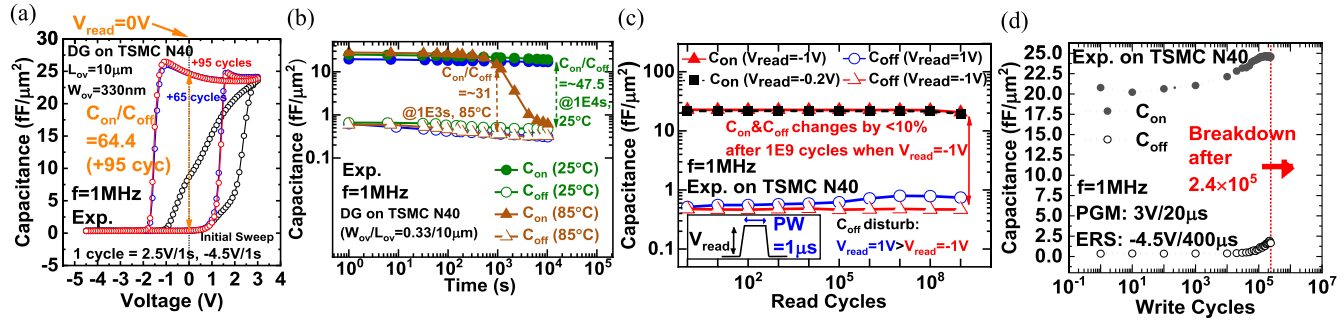


Fig. 7. Characterization results of optimized DG nvCAP on a TSMC N40 CMOS chip (W_{ov} = 330 nm L_{ov} = 10 μm, W/L/P = 20/10/5 μm). (a) *C–V* characteristics, (b) retention characteristics, (c) read-disturbance characteristics, and (d) write-endurance characteristics.

TABLE I
BENCHMARK OF DG NVCAP AGAINST BEOL-COMPATIBLE NVCAPS IN THE LITERATURE

	This Work	[5]	[16]	[17]
BEOL-integrated?	Yes	No	No	No
Substrate	TSMC N40 CMOS	Si	Si	Si
C_{on}/C_{off} ratio @ V_{read}	64.4 @0V	24 @0V	3.6 @-0.45V*	~1.3* @0V
CMW (fF/μm²) @ V_{read}	24.2 @0V	6.2 @0V	26.3 @-0.45V*	~8.5* @0V
AC small signal read frequency	1MHz	1MHz	10kHz	10kHz
Write Endurance	2.4E5	3E7	N/A	~1E7
Read Endurance @ V_{read}amplitude/ pulsewidth	>1E9 @1V,1μs	N/A	>1E9 @0.5V, 500ns	>1E11 @0.5V, 500ns
Retention (s)	1E3 @85°C >1E4 @25°C	1E3 @25°C	N/A	>5E4 @25°C

CMW: follows the definition in [17]
*Values calculated/read from the plots in the references

(Fig. 8). The role of the FEOL transistors in the 3T-1C unit cell in Cap-DCIM is to translate BEOL capacitance to a digital output bit corresponding to ~(IN × W). Here, IN (input bit) represents the amplitude of the input voltage pulse (V_{in}) applied to 3T-1C and W (weight bit) represents C_{on} or C_{off} of a BEOL-integrated DG nvCAP. C_{on} and C_{off} are written by applying a write voltage to V_{in}. It should be noted that the digital output bit (=OUT) of the 3T-1C unit cell is the complement of IN × W, unlike the conventional SRAM-based 10T DCIM. This issue is addressed by adopting a modified full adder [8] for the first stage of the adder tree in Cap-DCIM, which is indicated in the array-level description of Cap-DCIM [Fig. 9(a)–(c)]. The remaining stages are implemented using standard full adder tree structures [8]. The modified full adder in Cap-DCIM is designed to receive inverted inputs ($\bar{A}, \bar{B}$) but produce the same output bits (C_{out}, S) as the standard full adder. Multiplication results from 3T-1C unit cells are summed through the modified adder tree, eventually producing the same digital MAC output as the conventional SRAM-based 10T DCIM. As a result, Cap-DCIM achieves functional equivalence to the conventional SRAM-based 10T DCIM even though the output bit is the inverted form of IN × W. In this work, we adopt 36T design for the standard full adder and 38T design for the modified full adder [8]. It should be noted that there are various possible ways of implementing standard and modified full adders other than the designs adopted in

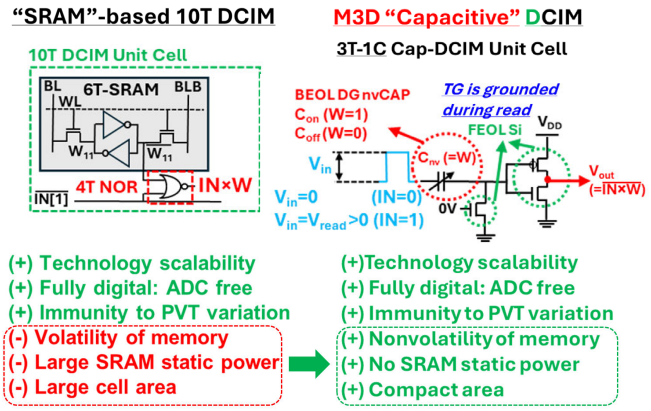


Fig. 8. Comparison of SRAM-based 10T DCIM and Cap-DCIM, showing that Cap-DCIM addresses the challenges of SRAM-based 10T DCIM.

this work. In addition, it is noteworthy that selectline (SL) of the Cap-DCIM array [Fig. 9(b)] is replaced with ground node (0 V) in this work for the proof-of-concept demonstration. However, in the practical array operation scheme, SL is biased accordingly so that negative write voltages are not required as V_{in}.

B. Validation of BEOL Capacitance Modulated FEOL Si Transistor Current Amplification

The operation of the 3T-1C unit cell is fundamentally based on the principle of BEOL capacitance-modulated FEOL Si current amplification [Fig. 10(a)]. This principle indicates that the C_{on}/C_{off} ratio of the BEOL-integrated DG nvCAP is amplified to a larger I_{on}/I_{off} ratio of the FEOL transistor. Large I_{on}/I_{off} ratios of the nMOS and pMOS transistors of the inverter in the 3T-1C unit cell fundamentally enable the digital voltage response in Cap-DCIM [Fig. 9(b)]. Here, I_{on}/I_{off} can be increased by increasing the ratios C_{on}/C_{int} and C_{int}/C_{off}. The measured C_{int} is 200 fF, which is dominated by the large IO pad parasitic capacitance at the top metal layer of the TSMC N40 CMOS chip. To achieve a high I_{on}/I_{off} ratio with NDRO, C_{on} and C_{off} of the BEOL-integrated DG nvCAP need to be optimized considering C_{int} and the NDRO condition.

For proof-of-concept, we experimentally validate this principle through a 2T-1C structure consisting of a BEOL-integrated DG nvCAP and FEOL N40 nMOS transistors in the FEOL. The demonstrated structure and the equivalent circuit model are described in Fig. 11(a) and (b). It should be noted that the "BE" of DG nvCAP is connected to the gate FEOL nMOS readout transistor during the TiN deposition/patterning step in the CMOS + X flow introduced in Fig. 6. Such a configuration is decided based on the single DG nvCAP read endurance measurement result in Fig. 7(c), indicating comparably lower read disturbance against negative V_{read} applied to BE. The two test setups (Test 1, Test 2) are described in Fig. 11(c). $I_{on}/I_{off} \sim 5441\times$ is achieved through Test 1 and $I_{on}/I_{off} > 200\times$ is achieved through Test 2 [Fig. 12(a) and (b)]. In Test 2 using minimum width core transistor as the read-out transistor, I_{off} is near or below the instrument limit, hindering the detection of the exact current value. Thus, we additionally demonstrate Test 1 using a large IO transistor for the readout transistor, which allows the detection of I_{off} and the large I_{on}/I_{off}.

Fig. 13 shows the operation region analysis of the DG nvCAP during its read operation under Test 2. An important observation from Fig. 13 is that NDRO is determined by the read of the PGM state (I_{on}), whereas the read of the ERS state (I_{off}) is nondestructive regardless of C_{off}, V_{read}, and C_{int}. For NDRO, the operation voltage shift (X_{on}) while reading I_{on} must stay within 1 V, which is the NDRO condition for the DG nvCAP, as verified from the single-device measurement [Fig. 7(c)]. A C_{on} target of 1800 fF in Test 2 leads to $X_{on} \sim 0.1 < 1$ V, allowing NDRO.

C. Modeling of BEOL Capacitance Modulated FEOL Si Transistor Current Amplification

Based on the experimental validation results of Cap-DCIM principles and the TSMC prelayout device model agreeing with the measured I_d–V_{gs} data after CMOS + X integration [8], we construct a compact model for the 2T-1C structure under Test 2 [Fig. 11(b)]. The experimentally validated 2T-1C model allows us to construct a reliable 3T-1C cell model for Cap-DCIM, which can then be used for circuit- and system-level evaluation of Cap-DCIM. C_{par} is the external parasitic capacitance component of C_{int}, originating from the BEOL interconnect and the top metal-layer IO pad. We model the DG nvCAP as a $1C_{nv}$–$1R_{nv}$ structure, where "C_{nv}" is C_{on} or C_{off} obtained from the AC small-signal capacitance measurement at 1 MHz [Fig. 12(b)] while "R_{nv}" represents the resistance originating from the nonoverlap region of the IWO channel, contact resistance between TE and IWO, and BEOL interconnect resistance, which cannot be captured through the AC small-signal capacitance measurement at a fixed frequency. In this work, R_{nv} is empirically fit with the measured transient current response of the 2T-1C structure, resulting in R_{nv} = 3.2 MΩ for the ON-state and R_{nv} = 4.5 MΩ for the OFF-state [Fig. 12(b)]. It will be an interesting research topic for future studies to further clarify the physical origin of R_{nv} by performing an impedance spectroscopy on the 2T-1C structure across a wide frequency range.

The simulated transient read-out current (I_N), I_{on}, and I_{off} under different operation conditions [input voltage pulse amplitude/width, ERS pulsewidth (PW)] in Fig. 14(a)–(c) show close agreement with the measurement results, indicating the validity of the 2T-1C compact model. It should be noted that the apparent delay in the peak response in the simulated I_N compared with the measured I_N is attributed to the settling time of the pulse measurement unit in the instrument (Keysight B1500A).

D. Functional Verification of Cap-DCIM

We construct a 3T-1C cell model for Cap-DCIM by incorporating TSMC's prelayout pMOS device model into the experimentally validated 2T-1C compact model. Based on the 3T-1C cell model, we validate the functional correctness of multiply-and-accumulate (MAC) operation of Cap-DCIM. Two 3T-1C unit cells are connected to the modified full adder, performing accumulation of the output bits [~(IN × W)] from

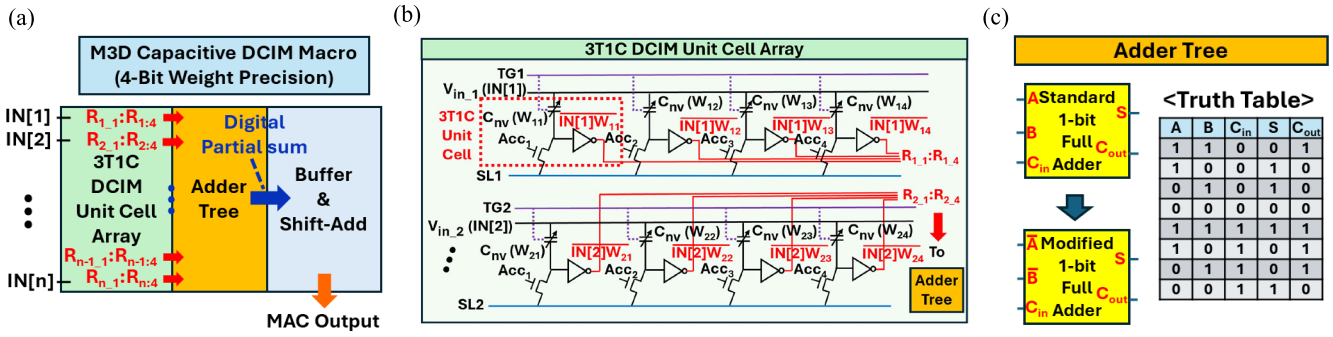


Fig. 9. (a) Schematic overview of the array-level (macro) implementation of cap-DCIM. (b) Description of the 3T-1C DCIM unit cell array in a Cap-DCIM macro. (c) Description of the modified full adder in the first stage of the adder tree design.

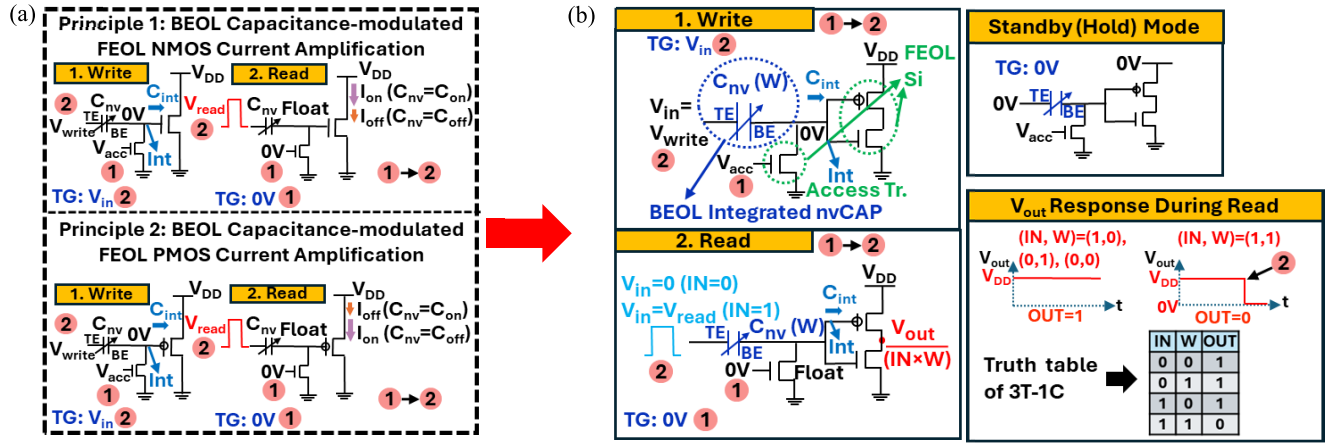


Fig. 10. (a) Description of the key principle of Cap-DCIM: BEOL capacitance-modulated FEOL Si current amplification. (b) Overview of the operation of the 3T-1C Cap-DCIM unit cell, including the write, read, and standby mode operation.

each 3T-1C unit cell. Fig. 15(a) and (b) shows the expected output waveform from the modified full adder, aligning with the truth table of the modified full adder presented in Fig. 9(c).

VII. Reliability Assessment of M3D Capacitive DCIM

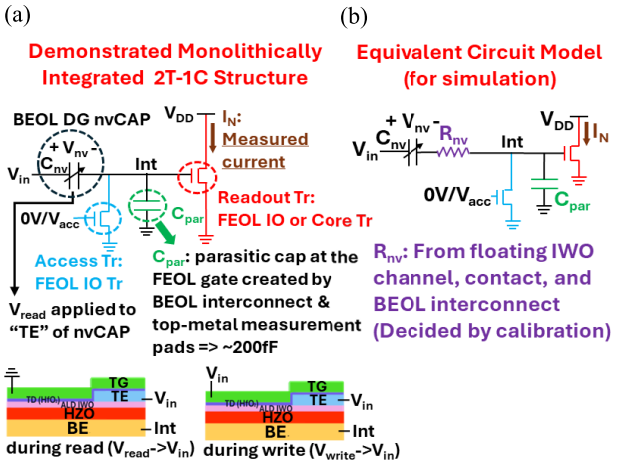
Reliability of the Cap-DCIM principle can be evaluated with the demonstrated 2T-1C structure. First, the retention characteristics are examined. At room temperature, the measured retention of the I_{on}/I_{off} ratio is $> 10^4$ s [Fig. 16(a)], consistent with the single device measurement results [Fig. 7(b)]. Next, the read endurance is examined. I_{on}/I_{off} ratio was able to be maintained after $> 10^7$ cycles of read voltage pulses with an amplitude of 1 V (V_{read} = 1 V) [Fig. 16(b)]. Altogether, the results highlight the nondestructive and nonvolatile nature of Cap-DCIM, which is a unique capability fundamentally unattainable in conventional SRAM-based DCIM designs.

It is noteworthy that a more rigorous reliability assessment should ultimately be conducted on the 3T-1C structure with BEOL-integrated nvCAPs and FEOL nMOS/pMOS transistors. Furthermore, in an array-level implementation, intercell capacitive coupling due to the toggling of the gate voltage of the access transistor, and intracell capacitive coupling similar to row hammer effects in DRAM may introduce additional reliability concerns. It will be worthwhile understanding the impact of these issues on the 3T-1C unit cell operation in future work. In addition, the write endurance of Cap-DCIM, governed by the reliability of both the FEOL access transistor and the BEOL-integrated DG nvCAP remains an important topic for future research.

VIII. Cell-Level Evaluation of M3D Capacitive DCIM

Based on the 3T-1C cell model (Section VI-B), a cell-level comparison between the conventional SRAM-based 10T DCIM and Cap-DCIM can be carried out at the 40-nm node. The notable benefit of Cap-DCIM is the substantial 111× static power reduction from the conventional 10T DCIM [Fig. 17(a)]. Such an improvement is enabled by replacing the SRAM cell (6T) consuming large static power with a DG nvCAP (1C). In the standby mode, Cap-DCIM cells experience only gate and junction leakage in the access transistor, unlike 10T DCIM cells, which additionally suffer from subthreshold leakage in multiple transistors. In addition, 3× cell area reduction can be achieved by reduced transistor count in Cap-DCIM [Fig. 17(b)].

Latency and dynamic energy consumption (at the ON-state) are evaluated as a function of C_{par} [external parasitic capacitance component of C_{int}, modeled in Fig. 11(b)]. The measured C_{par} is high (~200 fF) due to the existence of the parasitic capacitance originating from BEOL interconnects and the large



(c)

Test Type	Access Tr (Type, W/L, V_{acc})	Readout Tr (Type, W/L, V_{DD})	nvCAP Condition (P=5 μm)
Test1	IO, 91/0.55 μm, 2.5V	IO, 91/0.55 μm, 2.5V	W=20 μm L=10 μm L_{ov}=2.5 μm W_{ov}=0.6 μm
Test2	IO, 455/550 nm, 2.5V	Core, 120/40 nm, 1V	W=20 μm L=5 μm L_{ov}=2.5 μm W_{ov}=0.6 μm

Fig. 11. (a) 2T-1C structure, (b) equivalent circuit model for the demonstrated 2T-1C, and (c) test setups used for the demonstration.

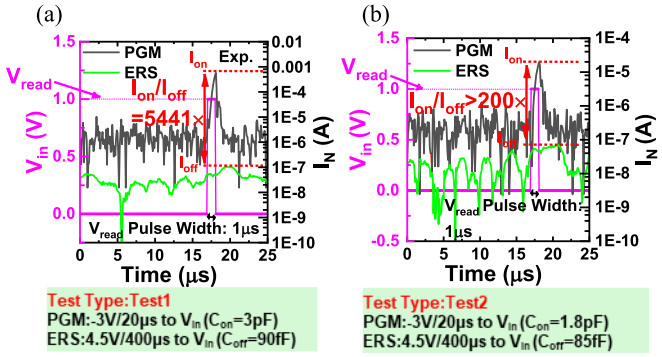


Fig. 12. Measured transient current (I_N) of the read-out transistor of the 2T-1C structure under the setups (a) Test 1 and (b) Test 2.

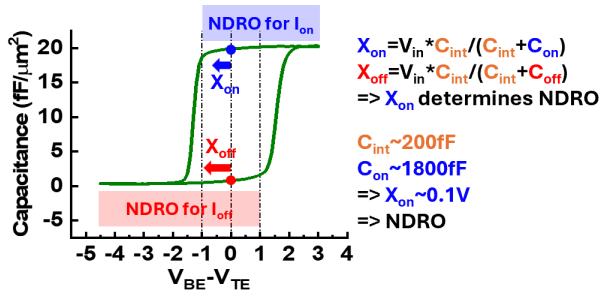


Fig. 13. Nondestructive read operation analysis of DG nvCAP during read in Cap-DCIM. Test 2 setup is taken as an example to demonstrate nondestructive read operation.

IO pad serving as the measurement pads. However, we assume that lower C_{par} ($\leq$ 1 fF) is achievable in advanced foundry processes by eliminating measurement pads and performing BEOL integration closer to FEOL.

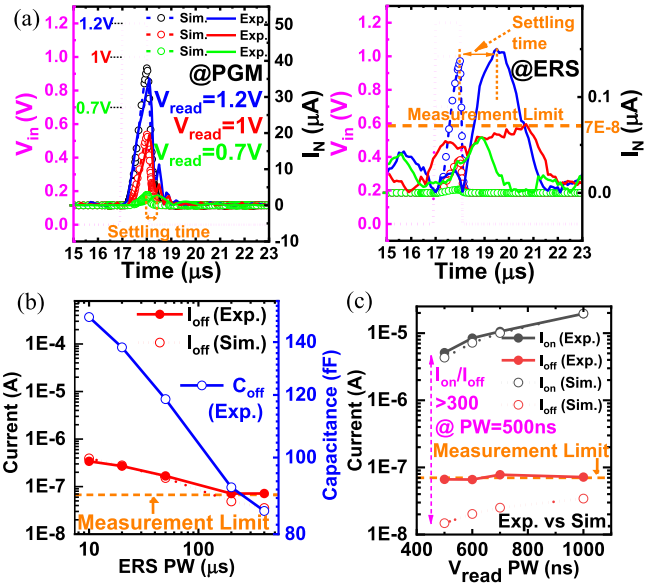


Fig. 14. Comparison of simulated I_N and measured I_N under Test 2. Comparison of (b) I_{OFF} and C_{OFF} under varying ERS PWs and (c) I_{ON} and I_{OFF} under varying V_{read} PWs between simulation and experiment under Test 2. For (b) and (c), V_{read} amplitude was 1 V. In (a), V_{read} PW was 1 μs. For write in (a)–(c), the PGM/ERS conditions in Fig. 12(b) were used.

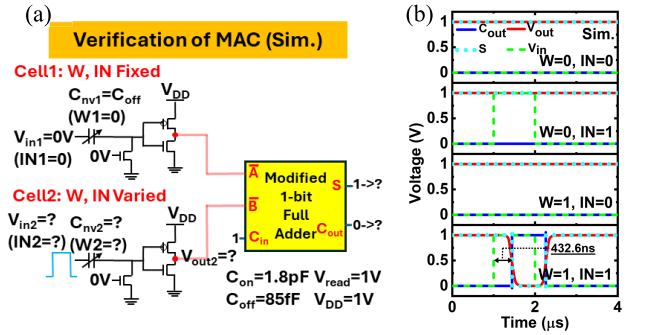


Fig. 15. (a) Simulated modified full adder + DCIM cell unit (with experimental dimension). (b) Simulated waveforms from the modified full adder + DCIM under different (IN2, W2) combinations.

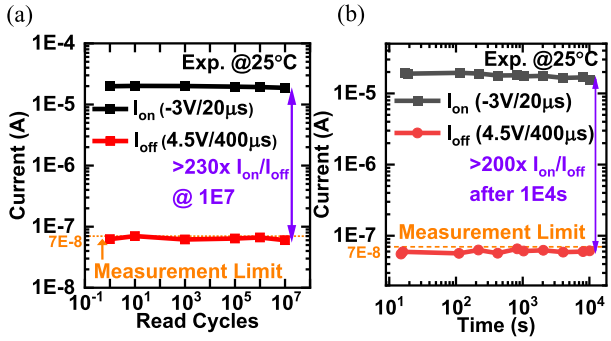


Fig. 16. Measured (a) read endurance and (b) retention characteristics of the proposed M3D Cap-DCIM under Test 2. In (a), a V_{read} pulsewidth of 1 μs and amplitude of 1 V were defined as one read cycle. The delay between consecutive read pulses was 1 μs. In (b), 2T-1C test structure was kept in standby mode, described in Fig. 10(b) Used PGM/ERS pulse conditions are specified in the legend.

Building upon these assumptions, we perform latency and dynamic energy evaluation at scaled C_{par} and $C_{on,target}$.

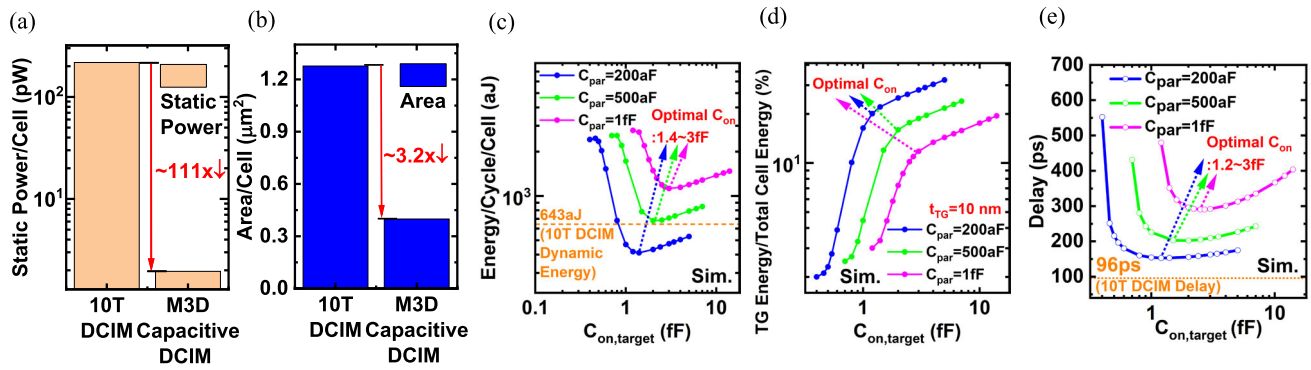


Fig. 17. (a) Static power and (b) area comparison between M3D Cap-DCIM and conventional SRAM-based 10T DCIM at the 40-nm node. (c) Dynamic energy per cycle of a 3T-1C cell. (d) Ratio of dynamic energy consumed by TG–TE parasitic capacitance to the total cell energy per cycle (assuming t_{TG} = 10 nm). (e) Read latency of a 3T-1C cell as a function of C_{par} at the 40-nm node.

Fig. 17(c)–(e) indicates that latency and dynamic energy consumption are dependent on C_{par}, suggesting C_{par} as an important parameter to be optimized in practical integration. $C_{par} \leq$ 500 aF and $C_{on,target}$ of 1.4–3-fF yield comparable latency and lower dynamic energy than the conventional SRAM-based 10T DCIM. To achieve the scaled $C_{on,target}$, we assume the experimentally demonstrated BEOL-integrated nvCAP with W/L/W_{ov}/L_{ov}/P = 20/10/0.6/2.5/5 μm [Fig. 12(a)] is scaled by a factor of $(C_{on,target}/3\ \text{pF})^{1/2}$ [3 pF is the ON-state capacitance of demonstrated BEOL-integrated DG nvCAP with W/L = 20/10 μm in Fig. 12(a)]. We present an improved energy estimation from the conference version of this work. Specifically, we consider that the parasitic capacitance between TG and TE ($A_{TG,TE}C_{TG,TE}$) additionally contributes a dynamic energy consumption of $\sim 1/2 A_{TG,TE}C_{TG,TE}V_{read}^2$ for each read operation. Here, $C_{TG,\ TE}$ is the normalized parasitic capacitance (~37.5 fF/μm^2 from measurement) between TG and TE and $A_{TG,TE}$ is the total overlap area between TG and TE, given as follows:

$$A_{TG,TE} = \left(W_{ov,target} + 2L_{ov,target}\right) \times t_{TG} \times (\#\ of\ TE\ electrodes)$$

where t_{TG} is the TG thickness, and $W_{ov,target}$ and $L_{ov,target}$ correspond to the overlap dimensions of the projected DG nvCAP achieving $C_{on,target}$. It should be noted that the top side area of the TE is not included in $A_{TG,TE}$ calculation, assuming TD deposited on top side of TE will be etched back in the practical integration scheme to expose TE for BEOL via connection.

To minimize the impact of $C_{TG,TE}$, we assume TG thickness is scaled down to 10 nm (from 40 nm in the experimental device) using the reported ALD technology in practical integration [18]. From Fig. 17(d), the total energy contribution from $A_{TG,TE}C_{TG,TE}$ is <25% at optimized $C_{on,target}$, indicating that the TG-related parasitics contribute relatively less to the total cell energy per cycle compared with other parasitic components and transistor switching. Further scaling of t_{TG} below 10 nm with ALD technology [18] is expected to minimize the energy contribution from TG-related parasitics even further.

Furthermore, the variability of the cell output with respect to C_{off} variation is examined for the voltage read-out (Cap-DCIM) and the current read-out (prior work [19]) in Fig. 18. It

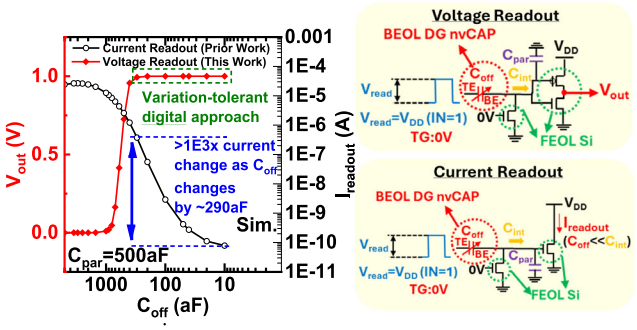


Fig. 18. Comparison of the voltage readout (this work) and current readout (prior work [19]) approach at the 40-nm node, highlighting the strong immunity of the voltage readout approach to C_{OFF} variation.

is indicated that C_{off} variation possibly due to PVT fluctuations and kT/C thermal noise can be effectively mitigated through M3D Cap-DCIM when compared with the current readout scheme [19].

IX. Array-Level Evaluation of M3D Capacitive DCIM

Finally, we address the crucial question—whether Cap-DCIM is truly beneficial at the array level. For the array-level simulation of Cap-DCIM, we use a modified version of NeuroSim V1.4, a CIM benchmark framework for leading edge technology nodes [6]. A deep neural network inference accelerator is constructed in NeuroSim V1.4, and the core performance metrics, energy efficiency, and compute density (TOPS/W, TOPS/mm^2) are derived from the simulated PPA results and are compared with nvCAP-based [5] and the state-of-the-art RRAM-based ACIM [20] approaches. ResNet-34 inference with the ImageNet dataset is used as a workload. The assumed ADC resolution for ACIM simulation is 7-bit, which is the required minimum precision to achieve software baseline accuracy [6]. The rest of the assumptions follow [5].

The key array-level benefit of Cap-DCIM is scalability, enabled by the fully digital operation of DCIM and BEOL integration of DG nvCAP. We project that Cap-DCIM can scale down to 3-nm technology node by adopting the 3-D Fin-nvCAP structure for the DG nvCAP, depicted in Fig. 19(a). Such a structure is fundamentally enabled by the

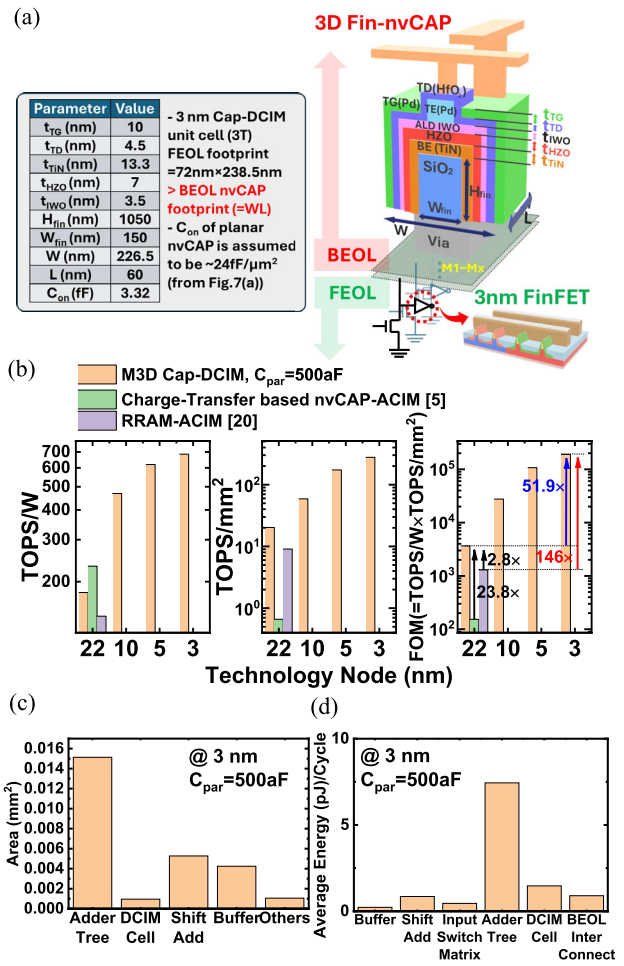


Fig. 19. (a) Description of the BEOL-integrated DG nvCAP with a 3-D Fin-nvCAP structure in 3-nm Cap-DCIM. (b) Array-level benchmark of Cap-DCIM, RRAM-based ACIM, and nvCAP-based ACIM. Array-level (c) area and (d) energy breakdown at 3-nm node.

ALD processes established for HfO_2 (TD), HZO, and IWO [5], which allow conformal deposition onto the 3-D fin structure. We additionally assume Pd is deposited with ALD technology for 3-D Fin-nvCAP at advanced technology nodes [18]. We note that the metal technology for TE and TG is not limited to Pd for practical implementation. The design parameters outlined in Fig. 19(a) ensure that the footprint of the BEOL-integrated DG nvCAP is smaller than the FEOL 3 T (access transistor + inverter) at the 3-nm node. This means that the cell area of Cap-DCIM is determined by the footprint of the FEOL 3 T. Moreover, the specified design values ensure $C_{on} > 3$ fF at the 3-nm node. For array-level simulation, C_{on}/C_{off} of 30 [from experimental data in Fig. 12(a) and (b)] is assumed. Furthermore, we assume C_{par} ($\leq$ 1 fF) is achievable at the 3-nm node, based on: 1) the specified capacitance per unit length (270 aF/μm) for the tightest metal pitch in the IRDS roadmap from the 14-nm down to 3-nm node and 2) publicly available interlayer dielectric (ILD) thickness of BEOL in leading-edge technology nodes, which scales with technology node. Based on publicly available cross-sectional data for a 14-nm Intel chip, the total ILD thickness from M0 to M9 is approximately 2.29 μm, and from M0 to M5 is approximately 655 nm [21]. Under these dimensions, one can assume parasitic capacitance of $C_{par} \leq 1$ fF at leading-edge nodes (e.g., 3 nm). It should be noted that the write voltage of the DG nvCAP must be scaled to remain within the maximum allowable V_{ds} overdrive of the given 3-nm technology, beyond which device reliability cannot be guaranteed. Recent works have reported 0.9-V write voltage for single-gate FeFETs [22] and 0.5-V write voltage for dual-gate FeFETs [23], which are comparable to the reported supply voltage of 3-nm technology (0.75 V) [24].

At the 22-nm node, Cap-DCIM exhibits ~ 23.8× and ~ 2.8× figure-of-merit (FOM = TOPS/W × TOPS/mm^2) improvements over nvCAP-based ACIM and RRAM-based ACIM, respectively. Technology scaling from the 22- to 3-nm node leads to a 51.9× FOM improvement for Cap-DCIM [Fig. 19(b)], demonstrating substantial performance advantages over ACIM approaches. As a result, 3-nm Cap-DCIM achieves ~ 146× FOM improvement compared with the state-of-the-art 22-nm RRAM-based ACIM.

It is well-established in the literature that in ACIM architectures, the ADC is the primary determinant of array-level and system-level performance [6]. On the other hand, in DCIM architectures, the adder tree predominantly governs array-level performance as indicated from the energy and area breakdown results at 3 nm shown in Fig. 19(c) and (d). In addition, the adder tree delay (~0.5 ns) accounts for ~83% of the critical delay of 3-nm Cap-DCIM. Therefore, further performance improvement of Cap-DCIM at leading-edge nodes will largely depend on the development of optimized adder-tree designs that effectively minimize the tradeoff between TOPS/W and TOPS/mm^2.

We emphasize that although this work concludes with an array-level evaluation of Cap-DCIM, practical neural network inference ultimately requires a full system-level implementation. At the system level, multiple arrays must be integrated with peripheral circuitry and global interconnects, including accumulation units, buffers, and H-tree networks, which inevitably introduce additional energy, area, and latency overhead. As a result, the achievable TOPS/W and TOPS/mm^2 at the system level are expected to degrade compared with the array-level metrics as shown in prior work [6]. Furthermore, the results will significantly vary depending on the adopted dataflow and detailed architectural choices. Building upon the device-level innovations and array-level scaling analysis presented in this study, future work will investigate system-level design tradeoffs of Cap-DCIM and benchmark its performance against existing computing paradigms under realistic architectural assumptions.

X. Conclusion

In this work, we have shown the first successful CMOS + X integration of a DG nvCAP on a foundry CMOS chip. We have demonstrated that the proposed DG nvCAP for "X" effectively mitigates the weak erase and poor retention challenges in oxide-channel FEs. Furthermore, we have elucidated the underlying physical mechanisms leading to these technical advancements. The optimized BEOL-integrated DG nvCAP on the TSMC N40 CMOS chip has achieved

a C_{on}/C_{off} of ~64.4 and NDRO of > 10^9 read-stress cycles at $|V_{read}|$ = 1 V. Furthermore, we have shown that BEOL-integrated DG nvCAP can be incorporated into DCIM to enable a Cap-DCIM paradigm, addressing the critical issues with ACIM and conventional SRAM-based 10T DCIM. We have experimentally validated and modeled the operational principle of Cap-DCIM, BEOL capacitance-modulated FEOL Si transistor current amplification through monolithically integrated DG nvCAP on the TSMC N40 CMOS chip. Finally, the projected 3-nm Cap-DCIM with 3-D Fin-nvCAP has achieved 146× higher figure-of-merit (TOPS/W × TOPS/mm^2) compared with leading ACIMs and 111× lower static power versus SRAM-based DCIMs, highlighting the strong promise of Cap-DCIM with technology scaling. The proposed Cap-DCIM, enabled by the BEOL-integrated DG nvCAP through the established CMOS + X integration flow, offers a promising pathway to overcome the long-standing scalability and variability limitations of leading ACIM approaches.

Acknowledgment

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